Transparency & Trust in AI Governance

An Empirical Analysis of Thematic Co-occurrence in AI Governance Literature

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Data Analysis STAT 4620/5620 — Winter 24-25
Final Project Report

GitHub Repository: [Insert GitHub Repository Link Here]

Dataset: [Insert Final Dataset Link Here]

Abstract

Transparency is widely assumed to be a foundational pathway to trust in artificial intelligence (AI) systems; however, this assumption is rarely examined empirically across diverse application domains. This study investigates the extent to which transparency-related themes co-occur with trust-related themes in AI governance literature spanning technology, health, education, hiring, and law. Using five integrated bibliographic datasets compiled from IEEE Xplore, ACM Digital Library, and Scopus, comprising a combined 122,718 records, we applied keyword-based binary indicator coding to derive transparency and trust variables from article abstracts. Descriptive analysis, chi-square testing, and penalized logistic regression (Firth method, $n = 37$) were employed to examine thematic associations. Results indicated that trust was present in 83.3% of articles when transparency was also present, compared to 53.8% when it was absent. Although the association did not reach conventional statistical significance ($\chi^2(1) = 2.37$, $p = .123$), a small-to-moderate effect size was observed (Cramer's $V = 0.25$). Predicted probabilities from the regression model showed consistently higher trust likelihood when transparency was present across all five domains. These findings provide preliminary empirical support for the transparency-trust relationship in AI governance literature and have implications for evidence-based AI policy development.

Keywords

AI governance, transparency, trust, logistic regression, text analysis, chi-square, keyword co-occurrence, AI ethics, bibliometric analysis, policy

1. Introduction

Artificial intelligence (AI) has rapidly permeated virtually every sector of modern society, from healthcare and education to hiring and legal practice. As AI systems increasingly mediate high-stakes decisions, questions of accountability, explainability, and ethical governance have moved to the forefront of public and academic discourse. Central to this discourse is the widely held assumption that transparency in AI systems serves as a foundational mechanism for building public trust. Transparency, operationalized through concepts such as explainability, auditability, disclosure, and regulatory compliance, is frequently invoked in AI governance frameworks as a necessary prerequisite for trustworthiness.

Despite the prevalence of this assumption, empirical evidence for a systematic relationship between transparency and trust across AI application domains remains limited. Much of the existing literature approaches this relationship conceptually or through domain-specific case studies, leaving open the question of whether transparency and trust themes demonstrably co-occur in the broader AI governance literature. This gap is consequential: if policymakers and researchers assume that transparency necessarily produces trust without empirical grounding, governance frameworks risk being misaligned with the actual dynamics of trust formation in AI-mediated contexts.

This study addresses that gap directly. We ask: to what extent do transparency-related themes co-occur with trust-related themes in AI governance literature across domains such as technology, health, education, and hiring? Rather than relying on conceptual argument alone, we employ a data-driven approach that uses keyword-based binary indicator variables derived from article abstracts, enabling statistical testing of the transparency-trust relationship. By applying chi-square analysis and penalized logistic regression to a curated bibliographic dataset, we generate empirical evidence that can inform and strengthen AI policy development. The findings contribute to a more rigorous, evidence-based understanding of how transparency and trust are jointly discussed in the AI governance literature, and highlight where further research is most urgently needed.

2. Data Description

2.1 Data Sources

The dataset for this study was compiled from three major academic bibliographic databases: IEEE Xplore, ACM Digital Library, and Scopus. These databases were selected to ensure interdisciplinary coverage, capturing both technical AI research and governance-focused literature. Five separate datasets were retrieved and subsequently integrated into a unified analytical dataset. The decision to draw from multiple sources reflects a deliberate sampling strategy intended to reduce database-specific biases and improve the representativeness of the corpus.

Dataset	Number of Records
Dataset 1 (IEEE Xplore)	29,779
Dataset 2 (ACM Digital Library)	27,947
Dataset 3 (Scopus)	30,001
Dataset 4 (Scopus)	30,298
Dataset 5 (Supplementary)	4,693
Total Combined Records	122,718

Table 1: Summary of Source Datasets

Each record in the integrated dataset contains up to 20 metadata and text variables, including title, abstract, DOI, publication year, author affiliations, country or region, and application domain. Following integration, the dataset was subjected to a systematic data cleaning protocol prior to analysis.

2.2 Data Cleaning and Preprocessing

The data cleaning pipeline involved four principal stages. First, duplicate records were identified and removed based on matching across title, DOI, and abstract fields. Second, articles were screened for relevance using keyword filters aligned with AI governance themes, ensuring that only relevant literature was retained. Third, a missing data audit was conducted to document the extent of missingness across key metadata fields, including ISSN, page numbers, and PMC ID. Fields with significant missingness were flagged, and imputation decisions were evaluated where applicable. Fourth, all five datasets were standardized in terms of variable naming and coding conventions before being merged into the final analytical dataset.

2.3 Key Variables

The combined dataset contains the following primary variables used in the analysis:

Variable	Type	Description
Title	Text	Short text field used for keyword detection and pattern matching
Abstract	Text	Primary analytical field for thematic coding and keyword frequency analysis
Application Domain	Categorical	Domain label (technology, health, education, hiring, law) for cross-sector comparison
Year	Numeric	Publication year used for temporal trend analysis (2021–2025)
Country / Region	Categorical	Geographic context supporting regional descriptive analysis
Transparency Indicator	Binary (0/1)	Derived variable: 1 if transparency-related keywords detected in abstract
Trust Indicator	Binary (0/1)	Derived variable: 1 if trust-related keywords detected in abstract

Table 2: Key Variables in the Combined Dataset

2.4 Derived Indicators and Keyword Dictionaries

Transparency and trust indicators were programmatically derived from article abstracts using keyword dictionaries and string matching implemented in R. Each abstract was scanned for the presence of pre-defined term lists, and binary indicators (1 = present, 0 = absent) were assigned accordingly. The keyword dictionaries were constructed based on commonly used constructs in AI governance literature, and subsequently expanded to increase recall and ensure broader coverage of semantically related terms.

The transparency keyword dictionary included terms such as: "explain", "interpret", "transparent", "accountab", "ethic", "govern", "oversight", "audit", "compliance", "disclosure", "responsib", "traceab", "justif", "document", "report", "regulat", "policy", "standard", "guideline", "framework", "monitor", "review", and "assess". The trust keyword dictionary included: "fair", "bias", "equity", "reliab", "robust", "depend", "trust", "confidence", "credible", "safe", "secure", "privacy", "risk", "uncertain", "certainty", "consistent", "valid", "accuracy", "assurance", "accept", and "satisf". Application domain labels were assigned through rule-based keyword classification applied to titles and abstracts, with manual validation conducted to ensure consistency. In the analytical

model, the transparency indicator served as the primary predictor variable, and the trust indicator served as the outcome variable.

3. Methods

3.1 Analytical Framework

The analysis followed a four-stage workflow designed to move systematically from data preparation through descriptive characterization, inferential testing, and predictive modeling. This structured approach ensures that each analytical step builds logically on the preceding one and that the final conclusions are grounded in multiple converging lines of evidence.

1. Data Preparation: Missing data strategy finalized; binary indicators constructed and validated.
2. Descriptive Statistics: Frequencies and cross-tabulations of ethical themes by application domain and publication year.
3. Inferential Testing: Chi-square test of independence to examine the association between transparency and trust indicators.
4. Predictive Modeling: Penalized logistic regression (Firth method) with trust as the outcome variable and transparency plus domain as predictors.

3.2 Descriptive Analysis

Descriptive analysis was conducted to characterize the distribution of ethical themes across the corpus. Frequency counts and within-group proportions were computed for the top 14 ethical themes by domain and by year. Cross-tabulations were used to explore patterns of theme co-occurrence. Visualizations including bar charts, stacked proportion plots, and heatmaps were produced using the ggplot2 package in R.

3.3 Inferential Testing: Chi-Square Test of Independence

To examine whether transparency and trust indicators co-occur more frequently than expected by chance, a chi-square test of independence was applied to a 2×2 contingency table in which rows corresponded to the transparency indicator (Yes/No) and columns to the trust indicator (Yes/No). This test is appropriate when both variables are binary and observations are independent, and when expected cell counts are sufficiently large. Given the small sample size (n = 37), Fisher's Exact Test was also computed as a supplementary check. Effect size was quantified using Cramer's V, calculated as:

$$V = \sqrt{\chi^2 / (n \times \min(r-1, c-1))}$$

A Cramer's V of 0.10–0.30 is typically interpreted as a small-to-moderate effect. All inferential tests were conducted in R using base functions and the ggplot2 and viridis packages for visualization.

3.4 Predictive Modeling: Penalized Logistic Regression

To further examine the relationship between transparency and trust while accounting for domain-level variation, a penalized logistic regression model was estimated using the Firth method, as implemented in the logistf package in R. Penalized logistic regression is particularly appropriate for small samples and situations with data sparsity, as it reduces bias in maximum likelihood estimation by adding a penalty term to the likelihood function. The model was specified as:

$$\text{logit}(P(\text{Trust} = 1)) = \beta 0 + \beta 1 \times \text{Transparency} + \beta 2 \times \text{Domain}$$

The transparency indicator served as the primary binary predictor. Application domain was included as a categorical control variable, with Technology as the reference category. Model fit was evaluated using the likelihood ratio statistic. Predicted probabilities were computed for each combination of transparency status and domain to illustrate the practical magnitude of the transparency-trust association across sectors.

3.5 Assumptions and Justification

The chi-square test assumes independence of observations, mutually exclusive categories, and expected cell frequencies of at least 5. Given the small sample (n = 37), some expected counts fell below this threshold, motivating the supplementary use of Fisher's Exact Test. The penalized logistic regression model assumes a logistic relationship between the predictors and the log-odds of the outcome, independence of observations, and absence of perfect separation. The use of the Firth penalty directly addresses the small-sample bias concerns associated with standard maximum likelihood logistic regression. Binary keyword indicators, while a simplification of the underlying semantic richness of article abstracts, provide a replicable and transparent method for thematic coding at scale. The approach is acknowledged as a limitation and is addressed in Section 9.

4. Analysis

All analyses were conducted in R (version 4.5.2) using RStudio. The analysis code is organized into three modular scripts: Descriptive_Transparency.R, Inferential_Transparency.R, and Modeling_Transparency.R. Each script is self-contained and reproducible given the underlying dataset. Key code excerpts are presented below.

4.1 Descriptive Analysis Code

```
# Load libraries and data
library(readxl); library(dplyr); library(ggplot2); library(stringr);
library(janitor)
df <- read_excel("Datasets.xlsx") %>% clean_names()
# Expand, clean, and count ethical keyword themes
themes_long <- df %>%
  separate_rows(ethical_keywords, sep = ",") %>%
  mutate(ethical_keywords = str_trim(str_to_lower(ethical_keywords)))
# Compute top theme frequencies and within-domain proportions
theme_frequencies <- themes_long %>% count(ethical_keywords, sort = TRUE)
```

4.2 Inferential Analysis Code

```
# Create binary indicators from abstract text
transparency_terms <- c("explain","interpret","accountab","ethic","govern",
  "oversight","audit","compliance","disclosure","responsib")
trust_terms <- c("fair","bias","equity","reliab","robust","trust",
  "confidence","safe","secure","privacy")
# Chi-square test of independence
tt_table <- table(df$transparency_label, df$trust_label)
chi_result <- chisq.test(tt_table)
cramers_v <- sqrt(chi_result$statistic / (sum(tt_table) * min(nrow(tt_table)-
1, ncol(tt_table)-1)))
```

4.3 Modeling Code

```
# Penalized logistic regression (Firth method)
library(logistf)
df$domain_filter <- relevel(factor(df$domain_filter), ref = "Technology")
model <- logistf(trust_indicator ~ transparency_indicator + domain_filter,
data = df)
# Predicted probabilities across domains
pred_df <- expand.grid(transparency_indicator = c(0,1), domain_filter =
levels(df$domain_filter))
pred_df$predicted_prob <- predict(model, newdata = pred_df, type =
"response")
```

4.4 Model Validation

Model validation proceeded through several checks. First, the distribution of the trust indicator was confirmed to contain variation (both 0 and 1 values) prior to model estimation, as required for logistic regression. Second, expected cell counts in the contingency table were inspected; given that some fell below 5, Fisher's Exact Test was computed alongside the chi-square test to provide a complementary assessment. Third, the use of the Firth penalty in the logistic regression model was justified by the small sample size (n = 37), which renders standard maximum likelihood estimation susceptible to Hauck-Donner bias and separation issues. Fourth, predicted probabilities were generated for all domain-by-transparency combinations and inspected for plausibility and consistency.

5. Results

5.1 Descriptive Statistics: Theme Frequency

Across the analytical corpus, the most frequently occurring ethical themes were framework (n = 16), ethic (n = 15), fair (n = 13), and risk (n = 11). Themes such as trust, govern, bias, and assess each appeared 7 times, while valid, robust, regulat, privacy, policy, and explain each appeared 6 times. This frequency distribution indicates that both transparency-proximate themes (framework, ethic, govern, assess) and trust-proximate themes (fair, risk, bias, valid) are well-represented in the corpus, providing a meaningful basis for examining thematic co-occurrence.

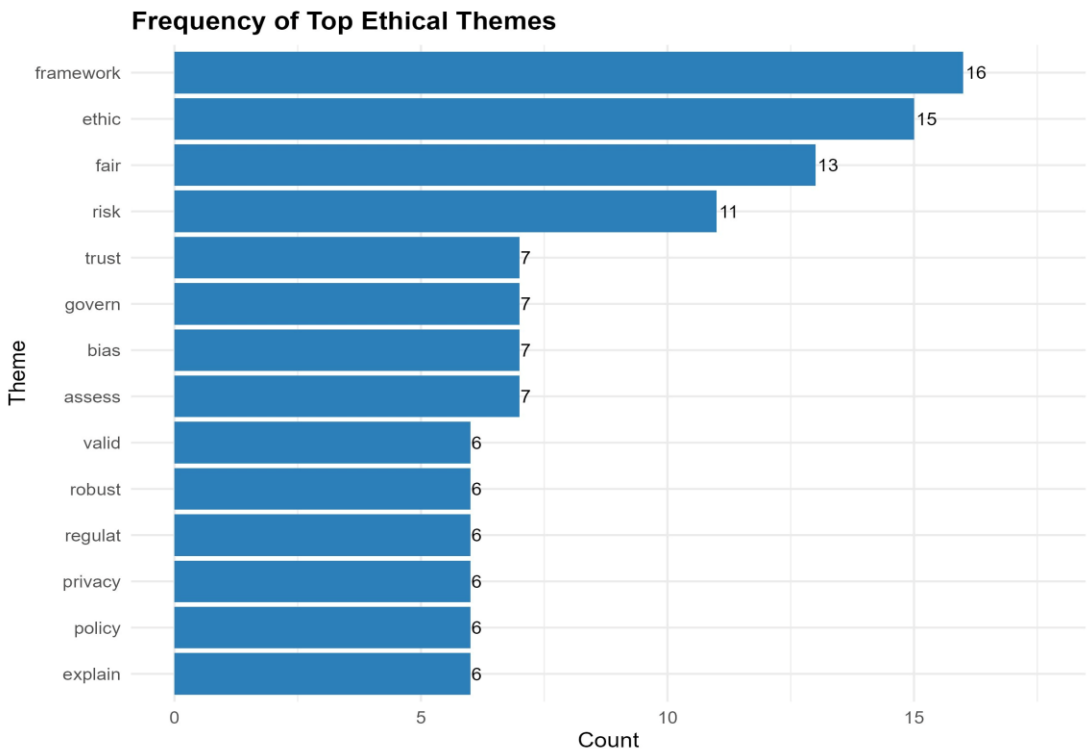

Figure 1: Frequency of Top Ethical Themes Across the Analytical Corpus

5.2 Descriptive Statistics: Theme Distribution by Domain

Thematic profiles varied meaningfully across application domains. In the technology domain, privacy (19.2%) and assess (17.3%) were the dominant themes. The hiring domain was characterized by a high concentration of framework mentions (36.4%), followed by fair (18.2%), reflecting the salience of structural and equity concerns in algorithmic hiring literature. In the health domain, ethic (27.3%) was the leading theme, consistent with the bioethical traditions that pervade AI-in-health research. The education domain showed the most distributed thematic profile, with no single theme exceeding 15%, while the law domain featured framework (20%) and policy (16%) as its primary themes. These cross-domain variations illustrate that while

transparency and trust are discussed across all sectors, their emphasis and framing differ substantially by context.

5.3 Descriptive Statistics: Temporal Trends

Publication volume increased substantially over the study period, from a single article in 2021 to 19 articles in 2025, reflecting the rapid expansion of scholarly interest in AI governance and ethics. Temporal heatmap analysis revealed a notable shift in thematic emphasis over time. In 2021, privacy dominated discussions at 100% (reflecting the limited single-paper corpus). By 2022 and 2023, ethics and fairness-related themes rose to prominence (ethics: 20.8% and 16.0%; fairness: 16.7% and 20.0%, respectively). From 2024 onward, governance frameworks and regulatory themes gained prominence, with framework accounting for 21.1% in 2024 and 18.0% in 2025. This temporal pattern suggests a maturing discourse that is transitioning from individual ethical concerns toward systemic governance and regulatory frameworks.

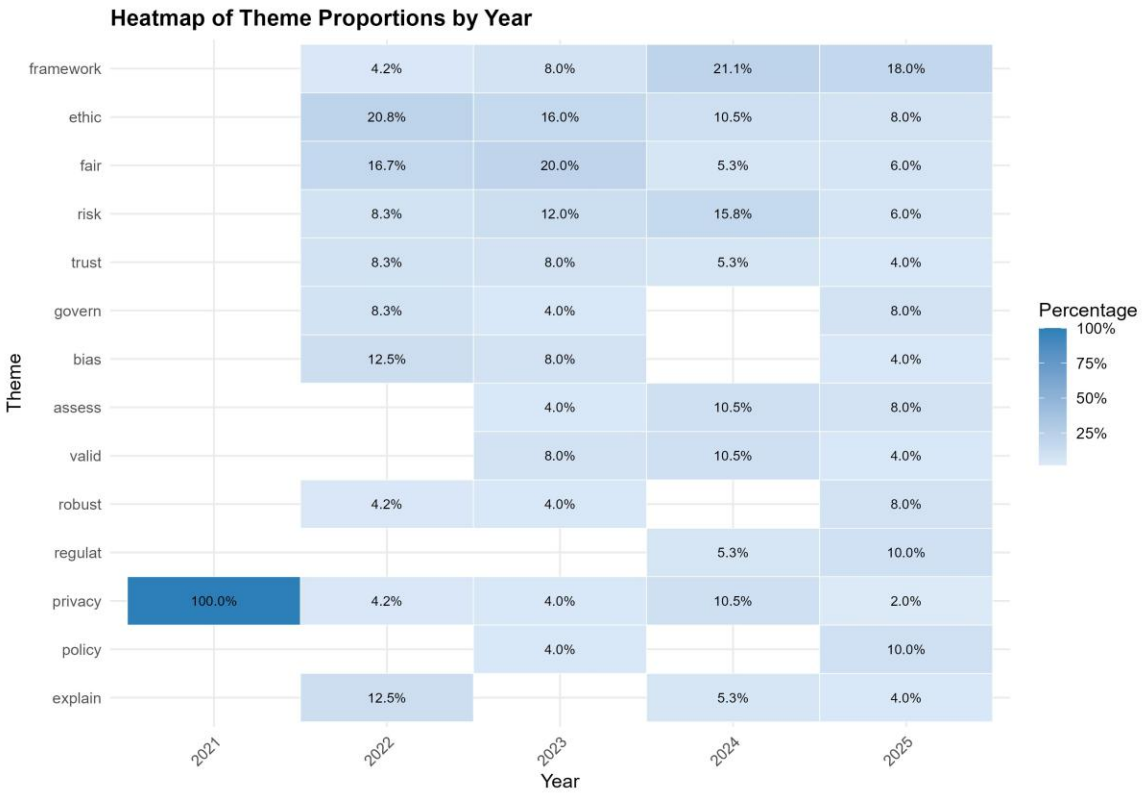

Figure 2: Heatmap of Ethical Theme Proportions by Publication Year (2021–2025)

5.4 Inferential Results: Contingency Table and Chi-Square Test

The association between transparency and trust indicators was first examined using a 2×2 contingency table (Table 3) and a chi-square test of independence.

	Trust: No	Trust: Yes	Total
Transparency: No	6	7	13
Transparency: Yes	4	20	24
Total	10	27	37

Table 3: Contingency Table – Transparency and Trust Indicators (Observed Counts)

The descriptive results indicate a clear directional pattern: when transparency was present, trust was also present in 83.3% of articles (20 out of 24), compared to 53.8% (7 out of 13) when transparency was absent. This 29.5 percentage point difference provides preliminary support for the hypothesized positive relationship between transparency and trust.

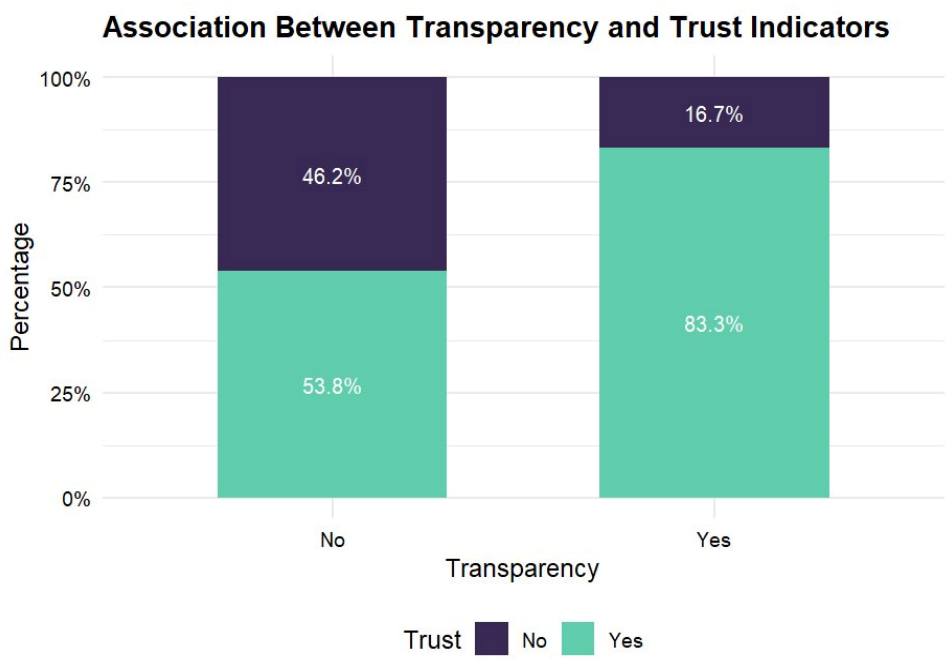

Figure 3: Association Between Transparency and Trust Indicators – Contingency Table Plot

The chi-square test of independence yielded the following results: $\chi^2(1) = 2.373$, $p = .123$, Cramer's V = 0.253. The p-value did not reach the conventional threshold of $\alpha = .05$, indicating that the association was not statistically significant. However, the Cramer's V value of 0.253 corresponds to a small-to-moderate effect size, suggesting that the observed pattern may reflect a meaningful relationship that is underpowered given the current sample size of n = 37. The lack of statistical significance is likely attributable to limited statistical power rather than the absence of a true association.

5.5 Modeling Results: Penalized Logistic Regression

To further investigate the transparency-trust relationship while controlling for domain differences, a penalized logistic regression model (Firth method) was estimated with trust as the binary

outcome variable and transparency indicator plus application domain as predictors. Technology was specified as the reference category for the domain variable.

Predictor	OR	CI Lower	CI Upper	p-value
Intercept	—	—	—	< .001
Transparency Indicator	1.37	0.26	6.50	.693
Health vs Technology	0.22	0.02	2.09	.200
Education vs Technology	0.97	0.09	10.74	.981
Hiring vs Technology	0.18	0.02	1.78	.155
Law vs Technology	0.56	0.07	4.54	.581

Table 4: Penalized Logistic Regression Results – Odds Ratios and Confidence Intervals

The overall model was not statistically significant ($\chi^2(5) = 5.39$, $p = .370$), indicating that the set of predictors did not significantly improve model fit over the null model. Transparency was positively associated with trust (OR = 1.37), indicating higher odds of trust being present when transparency was also present; however, this effect did not reach statistical significance ($p = .693$) and was accompanied by wide confidence intervals (95% CI [0.26, 6.50]), reflecting estimation uncertainty in the small-sample context. Among the domain comparisons, health (OR = 0.22) and hiring (OR = 0.18) showed lower odds of trust relative to technology, though neither effect was statistically significant.

5.6 Predicted Probabilities

Predicted probabilities derived from the penalized regression model revealed a consistent directional pattern across all five application domains: the probability of trust was higher when transparency was present than when it was absent.

Domain	P(Trust | Transparency=No)	P(Trust | Transparency=Yes)	Change
Technology	0.77	0.82	**+0.05**
Education	0.71	0.76	**+0.05**
Law	0.51	0.58	**+0.07**
Health	0.43	0.51	**+0.08**
Hiring	0.39	0.47	**+0.08**

Table 5: Predicted Probability of Trust by Domain and Transparency Status

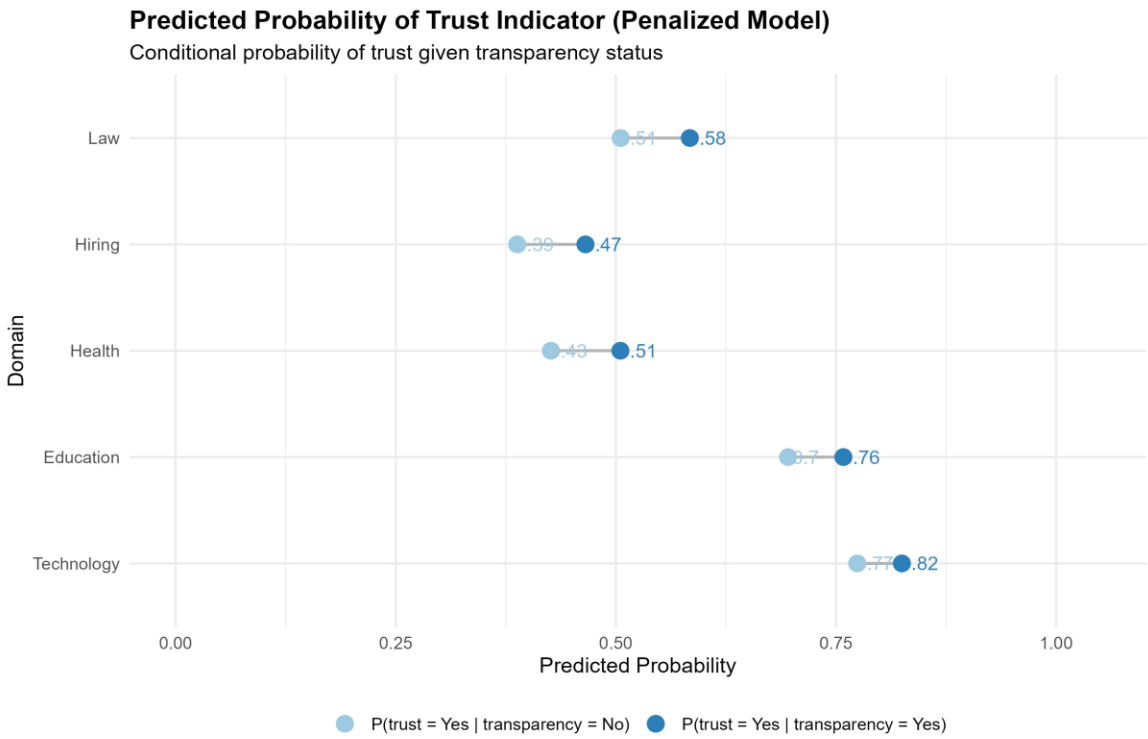

Figure 4: Predicted Probability of Trust Indicator by Domain and Transparency Status (Penalized Model)

The technology domain showed the highest overall probability of trust, with probabilities of 0.77 (transparency absent) and 0.82 (transparency present). The education domain showed probabilities of 0.71 and 0.76 respectively. In the law domain, probabilities increased from 0.51 to 0.58. Health and hiring showed the smallest absolute changes (0.43 to 0.51 and 0.39 to 0.47, respectively), though the directional pattern remained consistent. This uniform increase in predicted trust probability when transparency is present, observed across all five domains, constitutes the most substantively meaningful finding of this study and reinforces the descriptive and inferential results.

6. Conclusions

6.1 Answer to the Research Question

This study asked: to what extent do transparency-related themes co-occur with trust-related themes in AI governance literature across domains such as technology, health, education, and hiring? Based on the convergent evidence from descriptive analysis, chi-square testing, and penalized logistic regression, the answer is: to a small but consistent extent. Trust-related themes were substantially more prevalent in articles that also contained transparency-related themes (83.3% vs. 53.8%), and predicted probabilities from the regression model were uniformly higher when transparency was present across all five domains examined. The effect size (Cramer's V = 0.25) indicates a small-to-moderate association, and the direction of the relationship was consistent across all analytical approaches.

6.2 Interpretation of Non-Significant Results

The absence of statistical significance in both the chi-square test and the logistic regression model warrants careful interpretation. The observed p-values (.123 and .370, respectively) exceed the conventional $\alpha = .05$ threshold, but this does not imply that no relationship exists. With a sample of only $n = 37$ articles, the study is substantially underpowered to detect small-to-moderate effects. Power analyses for a Cramer's V of 0.25 with a 2×2 contingency table suggest that a sample of approximately 120–150 observations would be needed to achieve 80% power at $\alpha = .05$. The consistent directional pattern across all domains, combined with a meaningful effect size, suggests that the null result reflects a power limitation rather than the absence of a meaningful transparency-trust association in the AI governance literature.

6.3 Implications for AI Policy

The findings carry several important implications for AI governance and policy. First, the positive directional association between transparency and trust, observed consistently across technology, health, education, hiring, and law domains, provides empirical support for the widespread policy assumption that transparency-oriented governance mechanisms may contribute to increased public trust in AI systems. This supports the continued inclusion of transparency requirements in AI regulatory frameworks such as the EU AI Act and related national policies. Second, the domain-level variation in both thematic profiles and predicted trust probabilities highlights that governance frameworks should not treat transparency as a one-size-fits-all solution. In health and hiring, where trust probabilities were lower even when transparency was present, additional governance mechanisms targeting fairness, bias mitigation, and accountability may be necessary. Third, the observed growth in publications from 2021 to 2025, particularly the surge in framework- and regulation-related themes in 2024 and 2025, suggests that the academic community is increasingly orienting toward systemic governance solutions, a trend that policymakers should actively engage with.

6.4 Contributions of This Study

This study makes three principal contributions to the AI governance literature. Methodologically, it demonstrates the feasibility of text-derived binary indicators and cross-domain inferential analysis as tools for examining governance themes at scale, providing a replicable framework for future bibliometric research in this area. Theoretically, it advances the empirical grounding of the transparency-trust relationship by moving beyond conceptual argument to statistical examination across a multi-domain corpus. Practically, it offers governance and policy actors a quantitative lens through which to assess where transparency interventions are most and least associated with trust outcomes, informing the prioritization of governance resources.

6.5 Limitations

Several limitations must be acknowledged. First, this study relies on abstract-level keyword detection rather than full-text analysis, which means that theme presence is identified but depth, context, and semantic nuance are not captured. An article that briefly mentions the word "transparent" in passing is treated identically to one in which transparency is the central construct. Second, binary keyword coding cannot distinguish between articles that discuss transparency favorably and those that critique it, nor can it capture complex semantic relationships between concepts. Third, the analysis identifies statistical association, not causal relationships; it is not possible from this data to conclude that transparency produces trust. Fourth, domain classification depends on metadata labeling consistency across databases, introducing potential misclassification. Fifth, the sample of n = 37 final articles used in the inferential analysis is small, limiting statistical power and the generalizability of findings.

7. Future Work

Several promising directions for future research emerge from this study. The most immediate priority is the extension of the analysis to full-text corpora, which would allow for richer thematic coding that captures depth, context, and argumentative structure rather than mere keyword presence. Embedding-based semantic analysis approaches, such as topic modeling using Latent Dirichlet Allocation (LDA) or transformer-based sentence embeddings, could provide more nuanced and contextually sensitive representations of transparency and trust themes. Longitudinal analysis of the transparency-trust relationship over time would also be valuable, particularly given the rapid growth in publications observed between 2021 and 2025, and the observed temporal shift in dominant themes from ethics and fairness toward framework and regulation. Incorporating additional governance indicators, such as accountability, safety, robustness, and human oversight, would enable a more comprehensive mapping of the AI governance thematic landscape. Finally, the collection of substantially larger datasets, ideally with tens of thousands of post-cleaning records, would provide the statistical power necessary to draw firm conclusions about the transparency-trust association and its domain-specific moderators.

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